

Oobleck Armor Arena

Design armor with a material that acts like a liquid until impact.

Win condition: Best target protection with least material and cleanest reset.

THE SCIENCE YOU ARE USING

Shear-thickening fluids. Oobleck is cornstarch and water. Press it slowly and it flows like a liquid; hit it fast and it stiffens like a solid. Fast force jams the suspended particles together so they resist. This is shear thickening, and real impact-protection materials use the same trick.

Material efficiency. More material protects better but costs and weighs more. You optimize a sealed oobleck pad to protect a target with the least material, then test it with a controlled press, not a messy drop.

HOW TO BUILD AND RUN IT

- 01 Mix and feel it.** Mix cornstarch and water until it pours slowly but resists a fast poke.
- 02 Seal a pad.** Double-bag a measured amount of oobleck to make a clean, sealed armor pad.
- 03 Protect the target.** Place the pad over the target. Press a flat lid on top with a sharp counted "1-2-3 SLAP" (fast). For contrast, also lean slowly for 5 seconds and watch the target sink. Fast press protects; slow lean does not. Never rest a weight on the pad.
- 04 Trim the material.** Reduce the oobleck while still protecting the target. Track how little you can use.
- 05 Reset clean.** Show a clean reset with no leaks. Cleanup is scored.

Safety: No eating. Double-bag pads. Clean floors immediately. Goggles for splash risk. Allergy: Corn allergy note. Use sealed teacher-prepared bags if needed.

MATERIALS

Cornstarch	bulk
Water	shared
Zip bags	per team
Target (coin stack or foam square)	per team
Rigid lid for the press	shared
Measuring cup and spoon	per team

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why): _____

The ONE variable we are testing: _____

Baseline result: _____

After redesign: _____

DEFEND YOUR DESIGN

What evidence made you change your design? _____

If you ran it again, what is the ONE thing you would change? _____

HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Protection result	40
Material efficiency	20
Explanation	20
Redesign	10
Cleanup	10
Total	100